DNSC 6330: Responsible Machine Learning*

Lecture 03: Algorithmic Bias Measurement

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Outline (2 hours 15 minutes)

1. Recap of Lecture 02
2. Legal Foundations of Algorithmic Discrimination
3. Taxonomy of Algorithmic Harms
4. Formal Fairness Definitions
5. Live Coding: Subgroup Evaluation
6. Homework
7. Conclusion

Recap: What Transparency Gave Us

Established in Lecture 02

- Transparency $\neq$ trust; it is a *diagnostic* tool.
- GLMs and GAMs are interpretable *by design*.
- Post-hoc methods (LIME, SHAP, counterfactuals) answer different questions with different failure modes.
- Explanation washing: explanations without governance.

Open questions from Lecture 02

- SHAP reveals *which features* drive a prediction.
- It does **not** reveal whether the model *systematically disadvantages* a protected group.

$\Rightarrow$ We need: bias measurement.

Bridge to Today

Transparency reveals mechanics useful for interpretation, explanation and contestability. Bias measurement reveals **patterns** that may constitute unlawful discrimination or unjust disparate impact.

Recap: COMPAS Live-Coding Results

Empirical pattern ($n \approx 7{,}000$)

$$\widehat{\mathrm{FPR}}_{\text{Black}} > \widehat{\mathrm{FPR}}_{\text{White}} \tag{1.1}$$

$$\widehat{\mathrm{FNR}}_{\text{White}} > \widehat{\mathrm{FNR}}_{\text{Black}} \tag{1.2}$$

Risk scores were approximately *calibrated* across groups, yet error-rate parity failed.

Unanswered governance question

- SHAP explained *individual* scores.
- Does the *overall model* produce outcomes that would trigger legal scrutiny?
- Which statistical tests should a regulator apply?

Today's Goal

Turn legal exposure into **measurable quantities**.

Why Law Matters for ML Practitioners

- ML systems in **credit, employment, housing, education, and criminal justice** are subject to U.S. anti-discrimination law.
- Legal frameworks define the vocabulary of harm: who can be harmed, what counts as harm, what remedies exist.

Key Statutes

- Civil Rights Act of 1964
- Title VII
- Fair Housing Act of 1968
- ECOA of 1974
- ADA of 1990
- Title IX of 1972

Example Protected Categories

- Race, color, national origin
- Sex
- Religion
- Disability
- Age
- Familial status

Example Enforcement Agencies

- EEOC
- CFPB
- HUD
- DOJ
- OCC / Federal Reserve

Two Legal Theories of Discrimination

Disparate Treatment

- **Intentional** differential treatment because of a protected attribute.
- Requires showing discriminatory *intent*.
- Key case: *McDonnell Douglas Corp. v. Green* (1973).
- ML relevance:
  1. directly using race or sex as explicit features or
  2. using proxies that directly encode protected status.

Two Legal Theories of Discrimination

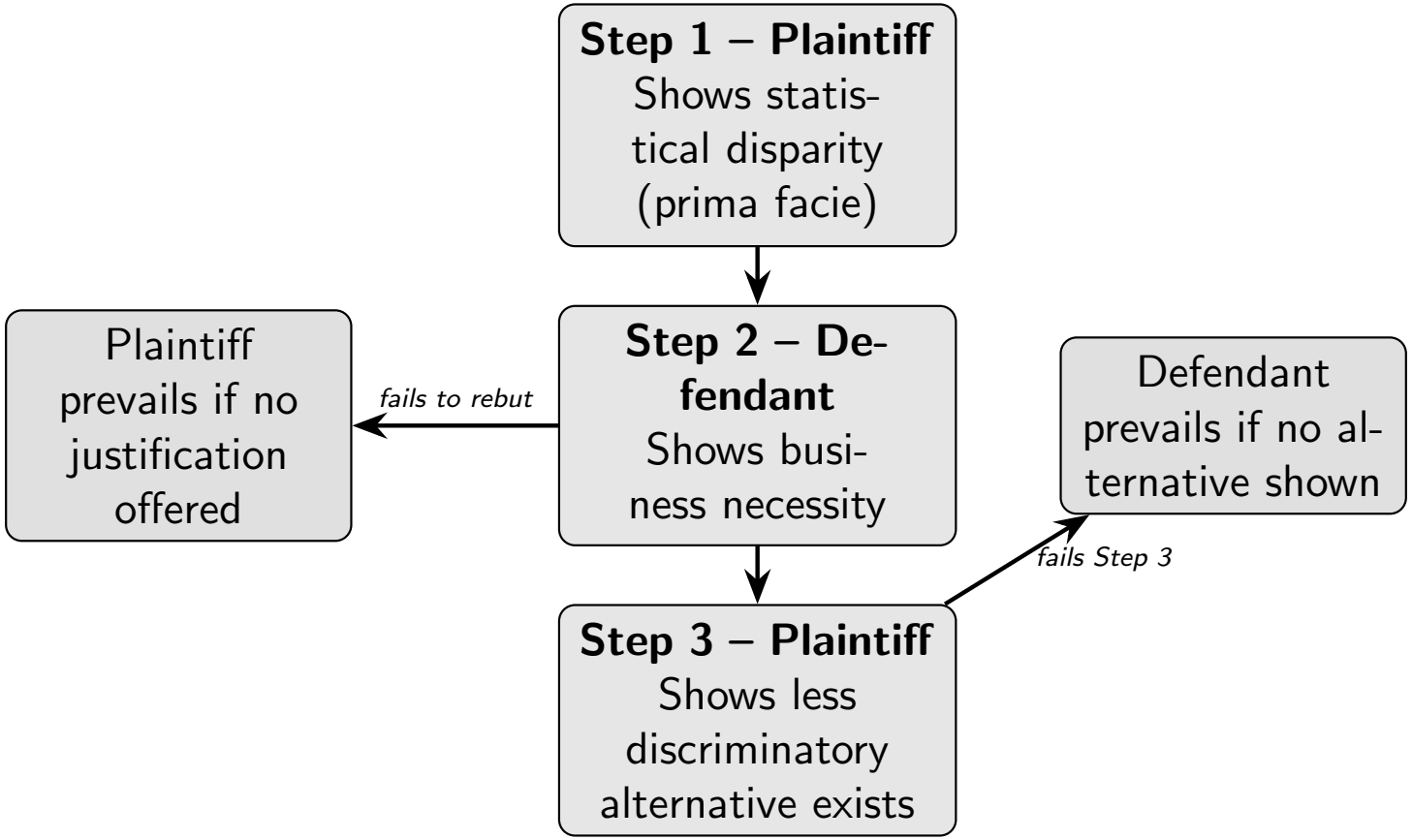
Disparate Impact

- A neutral-seeming policy that *disproportionately* harms a protected group, **regardless of intent**.
- Key case: *Griggs v. Duke Power Co.* (1971).
- ML relevance: for example, a model trained without race can still produce racially disparate outcomes through proxy discrimination.
- Business necessity defense available.

Core Implication for ML

Removing the protected attribute does **not** eliminate disparate impact. Proxies such as zip code, prior arrests, and credit score can re-introduce the protected signal.

The Burden-Shifting Framework



- Source: *Griggs v. Duke Power* (1971); *Watson v. Fort Worth Bank* (1988); N.J. DCR Guidance (2025).
- The 80% Rule (Adverse Impact Ratio) is the most common prima-facie statistical threshold used by the EEOC.

Regulating ML: How Old Law Applies to New Systems

Disparate Treatment via ML

- Using a protected attribute as a direct feature.
- Encoding protected status through high mutual information $\mathtt{I}$ proxies.
- Training data that reflects historical discrimination.

Disparate Impact via ML

- Aggregate metrics can mask the harm of concentrated subgroups.
- A 95% overall-accuracy model may have $\mathrm{FPR}_{\text{group A}} \gg \mathrm{FPR}_{\text{group B}}$.

Additional Regulatory Hooks

- **Explanation**: ECOA Sec. 701(d) requires adverse action notices.
- **Privacy**: demographic data needed for auditing creates tension with data minimization.
- **Consumer protection**: FTC Act Sec. 5.

Regulating ML: How Old Law Applies to New Systems

- **State-level**: N.J.A.C. 13:16 (2025); NYC Local Law 144 (2021).

Takeaway

U.S. antidiscrimination law may not require intent. Disparate outcomes can trigger liability.

N.J. DCR Guidance on Algorithmic Discrimination (2025)

- N.J. Division on Civil Rights issued formal guidance under the New Jersey Law Against Discrimination (NJLAD).
- Applies to any automated system affecting employment, housing, credit, or public accommodation.

Key Principles

- Disparate impact theory applies fully to algorithmic systems.
- Operators bear responsibility for auditing output, not just design.
- Using a vendor does not transfer liability.
- Intersectional harms are cognizable.

Operational Implications

- Document model purpose and scope.
- Conduct regular subgroup testing.
- Retain audit records.
- Build remediation workflows.
- Cannot delegate compliance to vendors.

Three Categories of Algorithmic Harm

Allocative Harms

Differential access to resources, opportunities, or services.

Examples: loan denials, hiring rejections, bail decisions, insurance pricing, ad targeting.

Legal hook: ECOA, FHA, Title VII.

Representational Harms

Stereotyping, erasure, or demeaning portrayals in ML outputs.

Examples: stereotyped image search results, word embeddings encoding bias, resume screening filtering names.

Legal hook: Hostile environment doctrine (Title VII), FHA.

Quality-of-Service Harms

Unequal predictive accuracy or system reliability across groups.

Examples: facial recognition failing on darker skin tones, speech-to-text errors for accented speech, medical risk models miscalibrated for minorities.

Legal hook: ADA, Title VI, FHA.

Proxy Discrimination

Proxy discrimination

Occurs when a model uses a feature X_j that is statistically associated with a protected attribute A, producing disparate outcomes even when A is excluded from the feature set.

Why it happens

- Training data encodes historical discrimination.
- Features like zip code, prior arrests, and school district carry high mutual information `I` with race.
- Formally: $I(X_j; A) > 0 \Rightarrow X_j$ is a potential proxy.

Measurement approach

1. Compute $I(\hat{Y}; A \mid X_j)$ for candidate proxies.

Proxy Discrimination

2. Test whether removing A reduces disparate impact: if $\mathrm{DI}(h_{-A}) \approx \mathrm{DI}(h)$, a proxy is active.
3. Examine SHAP values: high $|\phi_j|$ for known proxy features is a warning signal.

Proxy Discrimination

Interpreting the Disparity Test

"Fairness through blindness" fails. Removing the sensitive attribute rarely removes the sensitive *effect*.

Scenario	**What it means**
$DI(h_{-A}) \ll DI(h)$	Removing A reduces disparity $\rightarrow$ A was directly driving disparity
$DI(h_{-A}) \approx DI(h)$	Disparity persists $\rightarrow$ proxy features are carrying the signal
$DI(h_{-A}) \gg DI(h)$	Removing A worsens disparity $\rightarrow$ complex interaction / regularization effects

Intersectionality

Concept — Crenshaw 1989

Intersectionality recognizes that individuals belong to multiple protected groups simultaneously, and that discrimination can manifest at the *intersection* of identities in ways not captured by examining each dimension separately.

Intersectionality

Statistical implication

Testing $A =$ race and $A =$ sex separately **misses** harms concentrated on, e.g., Black women.

Facial recognition accuracy (Buolamwini & Gebru, 2018):

- White men: 0.8% error
- Black women: 34.7% error

Measurement practice

- Create interaction subgroups: $G = A_1 \times A_2$.
- Compute per-subgroup metrics: FPR_g, AIR_g for all $g \in G$.
- Report *worst-group* metric alongside aggregate.
- Always report n_g; small cells require caution.

Setup and Notation

Math Term	Meaning
$X \in \mathcal{X}$	feature vector
$\mathtt{A} \in \{0, 1\}$	protected attribute (0 = reference, 1 = non-reference)
$Y \in \{0, 1\}$	true outcome
$\hat{Y} = h(X) \in \{0, 1\}$	model prediction
$S = f(X) \in [0, 1]$	model score (continuous)

For group $A = a$:

$$\mathrm{TPR}_a = \mathbb{P}(\hat{Y} = 1 \mid Y = 1, A = a) \quad \mathrm{FPR}_a = \mathbb{P}(\hat{Y} = 1 \mid Y = 0, A = a)$$
$$\mathrm{FNR}_a = \mathbb{P}(\hat{Y} = 0 \mid Y = 1, A = a) \quad \mathrm{PPV}_a = \mathbb{P}(Y = 1 \mid \hat{Y} = 1, A = a)$$

Fairness Definitions: Group Fairness

Demographic Parity

$$\mathbb{P}(\hat{Y} = 1 \mid A = 0) = \mathbb{P}(\hat{Y} = 1 \mid A = 1)$$

Operationalized as the **Adverse Impact Ratio**:

$$\mathrm{AIR} = \frac{\mathbb{P}(\hat{Y} = 1 \mid A = 1)}{\mathbb{P}(\hat{Y} = 1 \mid A = 0)}$$

EEOC 80% Rule: $\mathrm{AIR} \geq 0.80$.

Equalized Odds — Hardt et al. 2016

$$\mathrm{TPR}_0 = \mathrm{TPR}_1 \ \wedge \ \mathrm{FPR}_0 = \mathrm{FPR}_1$$

Equal TPR *and* FPR across groups.

Fairness Definitions: Group Fairness

Equal Opportunity

$$\mathrm{TPR}_0 = \mathrm{TPR}_1$$

Equal recall across groups. A weaker condition than equalized odds.

Predictive Parity: Calibration

$$\mathbb{P}(Y = 1 \mid S = s, A = 0) = \mathbb{P}(Y = 1 \mid S = s, A = 1)$$

The score means the same across groups. Used as COMPAS's primary defense.

The Impossibility Theorem

Chouldechova (2017); Kleinberg et al. (2016)

When base rates differ between groups, **no** classifier can simultaneously satisfy:

1. Calibration (predictive parity),
2. Equal false positive rates, *and*
3. Equal false negative rates.

At most two of the three can hold simultaneously.

Why this matters

The ProPublica–Northpointe dispute about COMPAS was a dispute about *which* criterion to prioritize. Both sides were correct within their chosen criterion.

The Impossibility Theorem

The impossibility is not a technical flaw – it is a **value choice** that must be made explicitly and defended.

Governance Question

Which errors matter most, for whom, and under what legal standard?

Practitioners must *document* the choice and be prepared to justify it in court.

Individual vs. Group Fairness

Individual Fairness — Dwork et al. 2012

Similar individuals should receive similar predictions:

$$d_{\mathcal{X}}(x_i, x_j) \leq \varepsilon \;\Rightarrow\; d_{\mathcal{Y}}(h(x_i), h(x_j)) \leq \delta \tag{4.1}$$

Requires a *task-appropriate* similarity metric – hard to specify in practice.

Individual vs. Group Fairness

Tradeoffs

- Group fairness is *measurable* with demographic data but may sacrifice individual accuracy.
- Individual fairness is *conceptually appealing* but needs a contestable similarity metric.
- In regulated settings, group-level tests carry legal weight; individual fairness rarely does.

Discussion Prompt

A. Which fairness criterion should govern a recidivism risk score?

B. Should courts use group membership in calibration?

C. What does the Impossibility Theorem imply for policy?

Three Primary Discrimination Testing Metrics

Following Gill, Hall, Montgomery & Schmidt (2020) and the `solas-ai-disparity` Python library.

Adverse Impact Ratio

$$\mathrm{AIR} = \frac{\mathbb{P}(\hat{Y} = 1 \mid A = 1)}{\mathbb{P}(\hat{Y} = 1 \mid A = 0)} \tag{4.2}$$

- EEOC 80% Rule: $\mathrm{AIR} \geq 0.80$.
- Simple, long legal history.
- Sensitive to near-zero denominators.

Three Primary Discrimination Testing Metrics

Marginal Effect

$$\mathrm{ME} = \mathbb{P}(\hat{Y} = 1 \mid A = 1) - \mathbb{P}(\hat{Y} = 1 \mid A = 0) \tag{4.3}$$

- Absolute difference in selection rates.
- Less sensitive to small denominators.
- Analogous to absolute risk difference.

Three Primary Discrimination Testing Metrics

Standardized Mean Difference

$$\mathrm{SMD} = \frac{\bar{S}_1 - \bar{S}_0}{\sqrt{(\sigma_1^2 + \sigma_0^2)/2}} \tag{4.4}$$

- Works with continuous scores S.
- Cohen's d: < 0.2 small; ≥ 0.5 medium; ≥ 0.8 large.

Error-Rate Based Metrics

False Positive Rate Parity

$$\Delta\mathrm{FPR} = \mathrm{FPR}_{A=1} - \mathrm{FPR}_{A=0} \tag{4.5}$$

$\Delta\mathrm{FPR} > 0$ means the non-reference group is *over-flagged* when the true outcome is negative.

False Negative Rate Parity

$$\Delta\mathrm{FNR} = \mathrm{FNR}_{A=1} - \mathrm{FNR}_{A=0} \tag{4.6}$$

$\Delta\mathrm{FNR} > 0$ means the non-reference group is *under-served* when the true outcome is positive.

Error-Rate Based Metrics

Calibration within groups

For each score level s:

$$\mathbb{P}(Y = 1 \mid S = s, A = 1) \approx \mathbb{P}(Y = 1 \mid S = s, A = 0) \tag{4.7}$$

Tests whether the score carries the same predictive meaning across groups.

Impossibility Reminder

You **cannot** jointly satisfy calibration and FPR/FNR parity when base rates differ. Choose and document.

Practical vs. Statistical Significance

Statistical significance

- Tests whether an observed disparity is consistent with zero under the null.
- With large n, trivially small disparities become significant.
- Common tests: Fisher's exact, two-proportion z-test.

Practical significance

- Effect size: AIR, ME, SMD.
- EEOC 80% Rule is a practical threshold.
- Regulatory guidance requires *both* a significant p-value *and* a meaningful effect size.
- Small p, large n, tiny Δ: usually not actionable.
- Large effect, small n: operationally concerning even if not significant.

Practical vs. Statistical Significance

Rule of Thumb (Hall et al., 2020)

Report *both* a statistical test and an effect size. Flag cases where AIR < 0.80 **or** $p < 0.05$ and explain why the other criterion was not triggered.

The Solas-AI Disparity Testing Library

About

- Python library implementing legally grounded disparity metrics.
- Developed for real-world compliance contexts.
- Key metrics: AIR, ME, SMD, error-rate disparities.
- Supports binary classifiers and continuous scores.

Installation

```
pip install solas-ai-disparity
```

The Solas-AI Disparity Testing Library

Key Outputs

- Selection rates per group
- AIR with 80% threshold flag
- Marginal effect with p-value
- SMD for continuous scores
- Intersectional subgroup tables

The Solas-AI Disparity Testing Library

Example usage: calculating AIR

GitHub: github.com/SolasAI/solas-ai-disparity

```
import solas_disparity as sd

air_disparity = sd.adverse_impact_ratio(
    group_data=df,
    protected_groups=['black', 'hispanic', 'asian', 'female'],
    reference_groups=['white', 'white', 'white', 'male'],
    group_categories=['race', 'race', 'race', 'sex'],
    outcome=df['Approved'],
    sample_weight=None,
    air_threshold=0.90,
    percent_difference_threshold=0.0,
)
```

Traditional Tests for Discrimination

Following Barocas, Hardt & Narayanan (2023), Chapter 3.

Audit Studies

Field experiments: send matched pairs (identical except for a protected-attribute signal) into real decision systems.

Famous examples:

- Bertrand & Mullainathan (2004) – resume names: White-sounding names received 50% more callbacks.
- Ayres & Siegelman (1995) – car bargaining: Black testers paid ∼$1,100 more.

Traditional Tests for Discrimination

Observational Tests

Use existing data to test for disparate outcomes.

- **Outcome tests**: compare positive rates.
- **Separation tests**: compare error rates (equalized odds).
- **Calibration tests**: compare score reliability.
- **Selective labels**: beware survivorship bias in historical decisions.

Testing Discrimination in Algorithmic Systems

Three categories of ML-specific harms (Barocas et al., 2023):

NLP / Text

- Sentiment analysis assigns more negative sentiment to African American English.
- Co-reference resolution associates "doctor" with male pronouns.
- Word embeddings encode occupational gender stereotypes.

Computer Vision

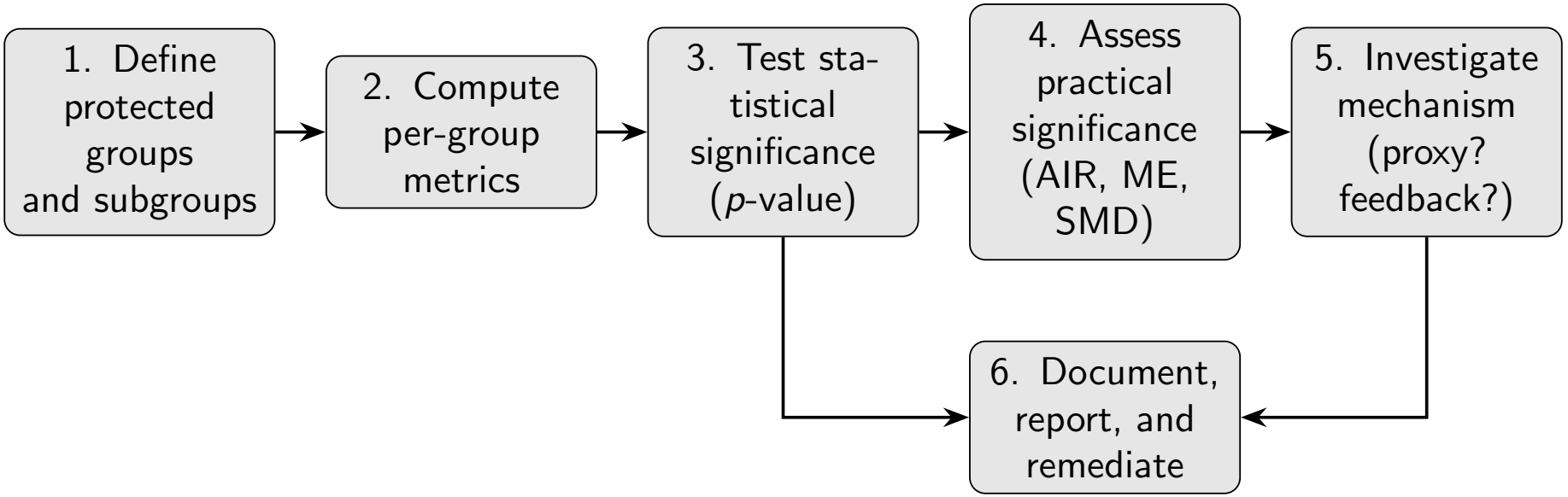
- Face recognition error rates vary by skin tone and gender (Gender Shades, 2018).
- Object detection mislabels people by race.
- Image captioning generates stereotyped descriptions.

Testing Discrimination in Algorithmic Systems

Search & Recommendation

- *Allocative*: fewer job ads shown to women.
- *Representational*: biased image results.
- *Quality of service*: worse recommendations for minority users.

A Systematic Framework for Bias Measurement



- Steps 3 and 4 are both required.
- Step 5 distinguishes diagnosis from explanation – essential for choosing the right mitigation strategy.
- Step 6 creates the audit trail a regulator or court would demand.

Discussion Prompt

Scenario

You are auditing a hiring algorithm for a large retailer. Your analysis reveals:

- Overall accuracy: 87%
- $\mathrm{AIR}_{\text{Black/White}} = 0.74$ below 80% threshold
- p-value for ME: $p = 0.31$ (not significant)
- $n = 420$ applicants; only 35 Black applicants

Discuss:

1. Is there evidence of disparate impact? Why or why not?
2. What is the role of sample size here?
3. What would you recommend the employer do?
4. How does the burden-shifting framework apply?

Live Coding: Session Plan

Continuing the COMPAS and HMDA capstone datasets from Lectures 01–02.

Learning Objectives

1. Compute AIR, ME, and SMD for the COMPAS logistic regression and gradient-boosted tree from Lecture 02.
2. Run error-rate disparity analysis (FPR, FNR) by race.
3. Compute intersectional subgroup metrics (race $\times$ sex).
4. Apply the `solas-ai` disparity library on the HMDA mortgage dataset (capstone preview).
5. Interpret results using the burden-shifting framework.

- Source code: [instructor GitHub link]
- Dataset: COMPAS Broward County cohort (Lecture 01 cleaning); 2024 HMDA LAR data (capstone).

Step 0: Setup and Imports

```
# Step 0: Setup and imports
import pandas as pd, numpy as np
from scipy import stats
import matplotlib.pyplot as plt
from statsmodels.stats.proportion import proportions_ztest
# pip install solas-ai-disparity
from solas_ai.disparity import DisparityTest

# Load and clean COMPAS data (Lecture 01 pipeline)

df = clean dataframe from lecture 02

print(f"N = {len(df):,}")
print(df[['race', 'sex', 'high_risk', 'two_year_recid']].head())
```

Step 1: Adverse Impact Ratio and Marginal Effect

```
# Step 1: AIR and Marginal Effect by race
def selection_rate(df, group_col, outcome_col, ref_group):
    """Selection rates, AIR, and ME relative to reference group
        ."""
    rates = (df.groupby(group_col)[outcome_col]
              .mean().rename('selection_rate').reset_index())
    ref_rate = rates.loc[rates[group_col]==ref_group,
                         'selection_rate'].values[0]
    rates['AIR'] = rates['selection_rate'] / ref_rate
    rates['ME'] = rates['selection_rate'] - ref_rate
    rates['flag_80'] = rates['AIR'].apply(
        lambda x: '*** BELOW 0.80' if x < 0.80 else '')
    return rates

sir = selection_rate(df, 'race', 'high_risk', ref_group='
    Caucasian')
```

Step 1: Adverse Impact Ratio and Marginal Effect

```
print(sir.sort_values('AIR').to_string(index=False))

# Two-proportion z-test: Black vs. White
groups = ['African-American', 'Caucasian']
ns = df[df['race'].isin(groups)].groupby('race')['high_risk'].
    count()
events = df[df['race'].isin(groups)].groupby('race')['high_risk
    '].sum()
stat, pval = proportions_ztest(events[groups].values, ns[groups].
    values)
print(f"\nAIR test: z = {stat:.3f}, p = {pval:.4f}")
```

Step 2: Error-Rate Disparity Analysis

```
# Step 2: FPR and FNR by race
def error_rates(df, group_col, pred_col, outcome_col):
    results = []
    for grp, g in df.groupby(group_col):
        tp = ((g[pred_col]==1) & (g[outcome_col]==1)).sum()
        tn = ((g[pred_col]==0) & (g[outcome_col]==0)).sum()
        fp = ((g[pred_col]==1) & (g[outcome_col]==0)).sum()
        fn = ((g[pred_col]==0) & (g[outcome_col]==1)).sum()
        results.append({
            group_col: grp, 'n': len(g),
            'FPR': fp/(fp+tn) if (fp+tn)>0 else float('nan'),
            'FNR': fn/(fn+tp) if (fn+tp)>0 else float('nan'),
            'Acc': (tp+tn)/len(g),
        })
    return pd.DataFrame(results)
```

Step 2: Error-Rate Disparity Analysis

```
er = error_rates(df, 'race', 'high_risk', 'two_year_recid')
print(er.sort_values('FPR', ascending=False).to_string(index=
    False))

# Highlight Black vs. White disparity
for grp in ['African-American', 'Caucasian']:
    row = er.loc[er['race'] == grp]
    print(f"{grp}: FPR={row['FPR'].values[0]:.3f} "
          f"FNR={row['FNR'].values[0]:.3f}")
```

Step 3: Standardized Mean Difference

```
# Step 3: SMD on continuous COMPAS decile score
def smd(df, group_col, score_col, ref_group):
    """Cohen's d vs. reference group."""
    ref = df.loc[df[group_col]==ref_group, score_col]
    results = []
    for grp, g in df.groupby(group_col):
        if grp == ref_group:
            continue
        sc = g[score_col]
        pooled = np.sqrt((ref.var() + sc.var()) / 2)
        d = (sc.mean() - ref.mean()) / pooled if pooled > 0 else
            0
        mag = ('small' if abs(d) < 0.2 else
              'medium' if abs(d) < 0.5 else
              'large' if abs(d) < 0.8 else 'very large')
        results.append({group_col: grp,
```

Step 3: Standardized Mean Difference

```
                        'mean_score': round(sc.mean(), 3),
                        'SMD': round(d, 3),
                        'magnitude': mag})
    return pd.DataFrame(results)

smd_tbl = smd(df, 'race', 'decile_score', ref_group='Caucasian')
print(smd_tbl.sort_values('SMD', ascending=False).to_string(index
    =False))
```

Step 4: Intersectional Subgroup Analysis

```
# Step 4: Intersectional analysis -- race x sex
df['subgroup'] = df['race'] + ' / ' + df['sex']

# Keep subgroups with n >= 30
counts = df['subgroup'].value_counts()
valid_sg = counts[counts >= 30].index
df_sub = df[df['subgroup'].isin(valid_sg)].copy()

sub_rates = (df_sub.groupby('subgroup')['high_risk']
            .agg(['mean','count'])
            .rename(columns={'mean':'selection_rate','count':'n
                '})
            .reset_index())

ref_rate = sub_rates.loc[sub_rates['subgroup']=='Caucasian / Male
    ',
```

Step 4: Intersectional Subgroup Analysis

```
                        'selection_rate'].values[0]
sub_rates['AIR'] = sub_rates['selection_rate'] / ref_rate
sub_rates['flag'] = sub_rates['AIR'].apply(
    lambda x: '*** BELOW 0.80' if x < 0.80 else '')

print(sub_rates.sort_values('AIR').to_string(index=False))

worst = sub_rates.loc[sub_rates['AIR'].idxmin()]
print(f"\nWorst: {worst['subgroup']} AIR={worst['AIR']:.3f} "
      f"n={worst['n']}")
```

Interpreting Solas-AI Output

Sample summary table

Group	AIR	ME	Flag
White (ref)	1.00	0.000	—
Asian	0.91	−0.031	—
Hispanic	0.76	−0.082	***
Black	0.71	−0.099	***

(Illustrative; run on actual data.)

Interpretation steps

1. Identify groups below AIR = 0.80.
2. Check p-value for statistical significance.
3. Assess effect size (ME, SMD): is it practically meaningful?
4. Trigger burden-shifting framework: business necessity?
5. Use SHAP (Lecture 02) to identify which features drive the disparity.

Interpreting Solas-AI Output

Connecting Lectures 01–03

L01 alignment gap → L02 transparency (SHAP, LIME) → L03 bias measurement (AIR, ME, SMD, error rates). These three form a **complete diagnostic pipeline**.

Visualizing Error-Rate Disparities

```
# Visualize FPR and FNR by race
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
er_plot = er[er['n'] >= 50].sort_values('FPR', ascending=False)
navy, red = '#00205B', '#B8192C'
colors = [red if r=='African-American' else navy for r in er_plot
    ['race']]

for ax, metric, title in [
        (axes[0], 'FPR', 'False Positive Rate by Race'),
        (axes[1], 'FNR', 'False Negative Rate by Race')]:
    ax.barh(er_plot['race'], er_plot[metric], color=colors)
    baseline = er_plot.loc[er_plot['race']=='Caucasian',
                           metric].values[0]
    ax.axvline(baseline, color='black', linestyle='--',
               label='Caucasian baseline', linewidth=1.5)
    ax.set_xlabel(metric)
```

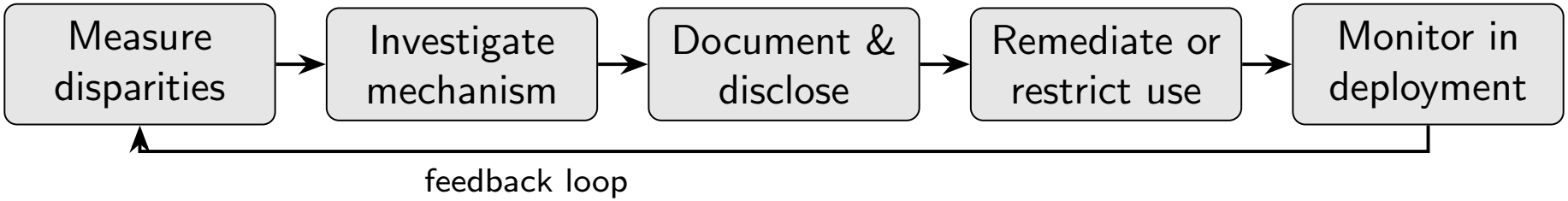
Visualizing Error-Rate Disparities

```
    ax.set_title(title)
    ax.legend(fontsize=8)

plt.tight_layout()
plt.savefig('error_rate_disparity.png', dpi=150, bbox_inches='
    tight')
plt.show()
```

From Measurement to Governance

Bias measurement is necessary but not sufficient for responsible ML.



What documentation should capture

- Reference group choice and justification
- Statistical tests and their assumptions
- Effect-size thresholds and their basis
- Whether business necessity was assessed
- Date, data version, and model version

Common remediation strategies

- Threshold adjustment per group
- Re-weighting training data
- Adversarial debiasing
- Feature removal or substitution
- Restrict model to lower-stakes contexts

Homework: Due at 12pm ET on April 6, 2026

Group Homework 3 – Mathematical

Let h be a binary classifier, $S \in [0,1]$ a score, $Y \in \{0,1\}$ the outcome, $A \in \{0,1\}$ a protected attribute.

a. State the conditions for *calibration*, *FPR parity*, and *FNR parity*.
b. Prove that when $\mathbb{P}(Y{=}1 \mid A{=}0) \neq \mathbb{P}(Y{=}1 \mid A{=}1)$, no non-trivial classifier satisfies all three simultaneously.
c. Apply the impossibility result to the COMPAS empirical findings from Lecture 01. Which criterion did COMPAS satisfy? Which did it violate?
d. Propose a governance rule: under what circumstances should an organization prioritize calibration over FPR parity? Justify using the burden-shifting framework.

Homework: Due at 12pm ET on April 6, 2026

Individual Homework 3 – Coding: Disparate Impact Audit

Using the cleaned COMPAS dataset and your model from Lecture 02:

1. Compute AIR, ME, and SMD for race and sex separately using the `solas-ai` Python library. Confirm both produce identical results.
2. Build an intersectional analysis (race $\times$ sex). Report the worst-group AIR and interpret it.
3. Compute FPR and FNR disparities by race. Test statistical significance with a two-proportion z-test.
4. Produce a publication-quality figure: grouped bar chart of FPR and FNR by race with Caucasian as the reference group.
5. Write a 300-word compliance memo addressed to a hypothetical regulator summarizing findings, metrics used, and limitations.

Submission: follow the same procedure as Homeworks 1 and 2.

Five Core Takeaways

1. **Disparate impact does not require intent.** A neutral model can produce discriminatory outcomes through proxy features that encode protected status.
2. **No single fairness metric is sufficient.** AIR, ME, SMD, FPR parity, and calibration each capture a different dimension. Use multiple and document your choices.
3. **The Impossibility Theorem is a value choice.** When base rates differ, you must choose which errors to minimize and be prepared to defend that choice legally and ethically.
4. **Intersectionality matters operationally.** Aggregate metrics can conceal concentrated harms at the intersection of race, sex, age, or disability.
5. **Measurement without governance is just a report.** Bias findings must feed into a documented audit trail, remediation workflow, and ongoing monitoring – not a one-time compliance checkbox.

Readings (for this lecture)

Required

1. Barocas, Hardt & Narayanan, *Fairness and Machine Learning* (2023), Chapters 3 and 6. fairmlbook.org.
2. Gill, Hall, Montgomery & Schmidt (2020). "A Responsible Machine Learning Workflow..." *Information*, 11(3), 137.
3. N.J. Division on Civil Rights (2025). *Guidance on Algorithmic Discrimination* and *Intro to Algorithmic Fairness*. njoag.gov.

Optional

1. Chouldechova (2017). "Fair Prediction with Disparate Impact." *Big Data*, 5(2), 153–163.
2. Buolamwini & Gebru (2018). "Gender Shades." *FAccT 2018*.
3. Crenshaw (1989). "Demarginalizing the Intersection of Race and Sex." *U. Chicago Legal Forum*.

Thank You!

Any questions?

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Office hours: Funger Hall 415R, Fri 5:00–7:00 PM

Appendix: Grading and Policy

Grading

- Weekly assignment: $\frac{2.5}{10}$
- Public GitHub with working codebase: $\frac{2.5}{10}$
- Weekly quiz: $\frac{2}{10}$
- Capstone: $\frac{3}{10}$

Coding / Scripting: Python, R

Collaborative Infrastructure: GitHub

Capstone Project

- 2024 HMDA Loan/Application Records (LAR) data.
- Goal: classify `action_taken`.
- Label: values 1, 2 $\rightarrow$ 1; value 3 $\rightarrow$ 0.
- Groups of 3–5 (randomly assigned by instructor).
- Office hours: Funger 415R, Fri 5:00–7:00 PM.
- Class resources: instructor materials and Patrick Hall's additional resources.